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Anatomy of a Modular Toolchain for Reproducible Bioinformatics Workflows

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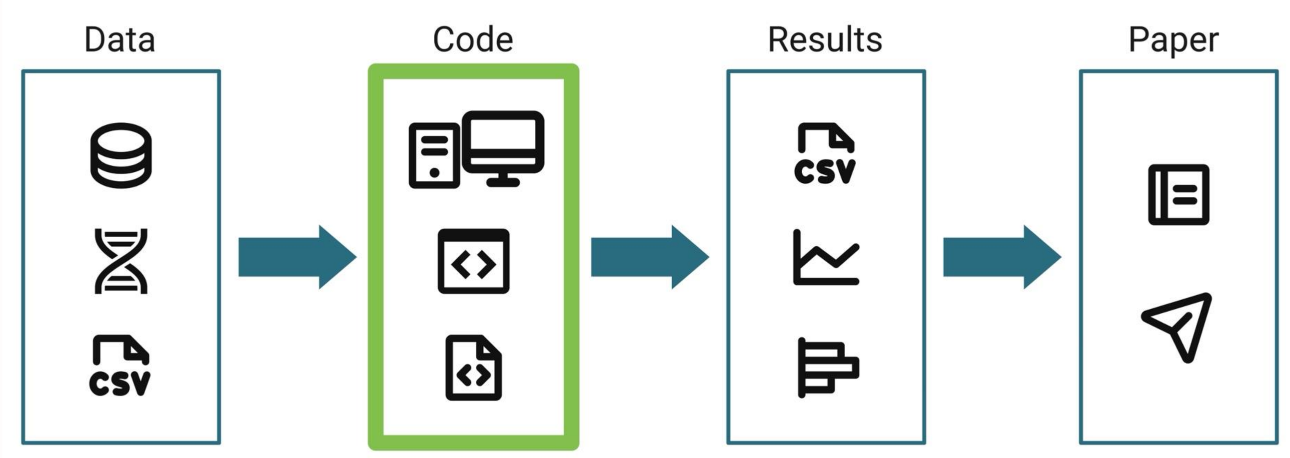
CCBR

INTRODUCTION

Reproducibility is the foundation of trustworthy data-intensive biological research. Our framework implements core reproducibility principles across our software suite.



REPRODUCIBILITY *Same data; same methods*	**REPLICABILITY** *Different data; same methods*
ROBUSTNESS *Same data; different methods*	**GENERALIZABILITY** *Different data; different methods*

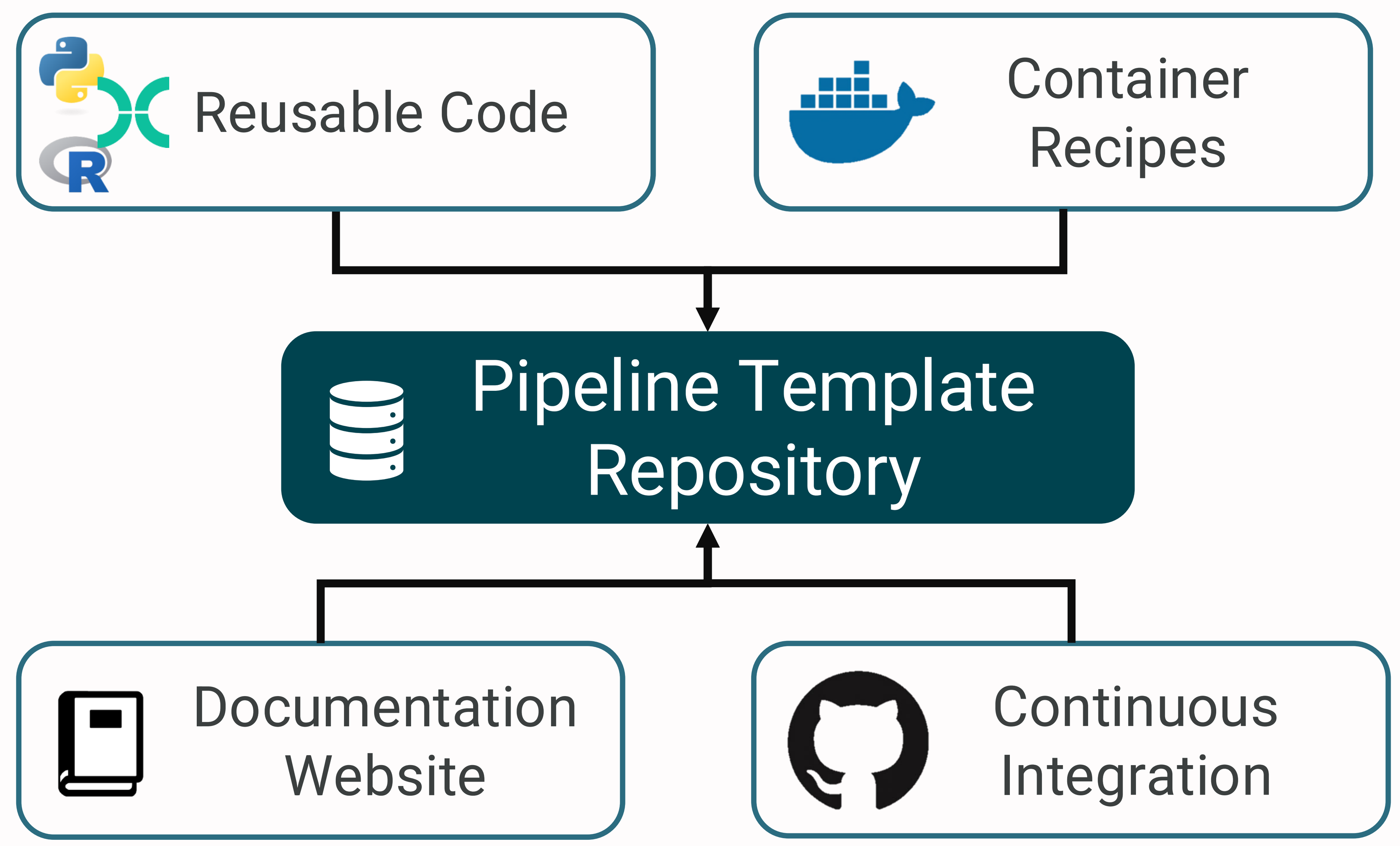
Schloss PD. 2018. 10.1128/mBio.00525-18.

BEST PRACTICES

- **Version Control**
- **Modular Design**
- **Automated Tests**
- **Documentation**

We enforce consistency through templates, modularity, and automation to prevent technical debt, streamline maintenance, and enable long-term sustainability.

TOOLCHAIN FOR REPRODUCIBLE BIOINFORMATICS



Unified framework from template to deployment with automated quality checks.

Fix once; deploy everywhere	Consistent CLI & Implementation	Automated testing & quality checks

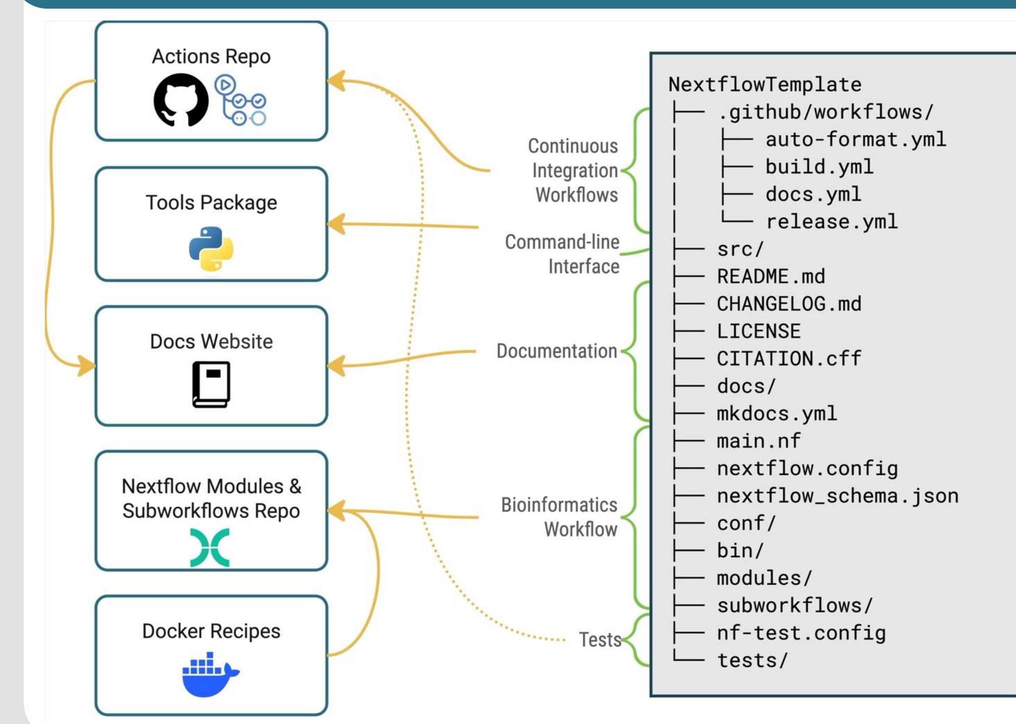
PIPELINES BUILT

ChIP-Seq **CHAMPAGNE**	Whole Genome **LOGAN**	scRNA-Seq **SINCLAIR**	CRISPR-Seq **CRISPIN**

10+ pipelines on NIH Biowulf HPC cluster

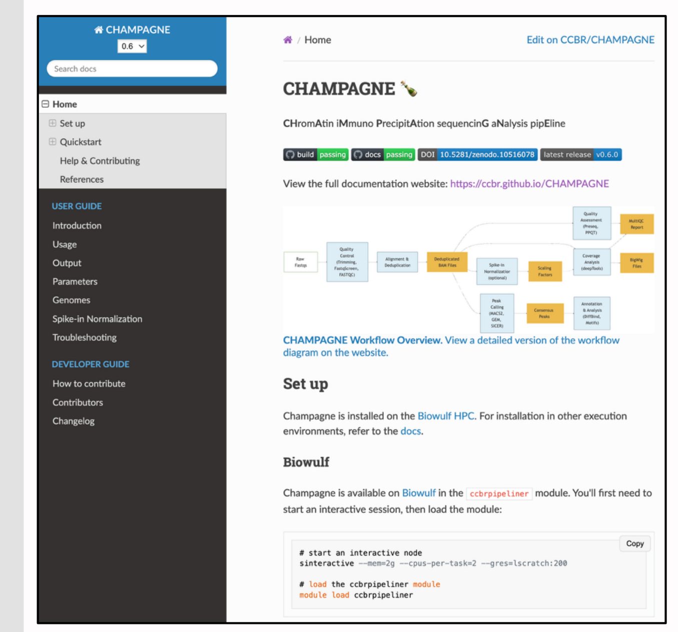
```
$ module load ccbrpipeliner
```

PIPELINE TEMPLATE



- Standardized directory structure for all pipelines
- Built in CI, testing, and containerization
- Easily extendible for newer methods and technologies

CLI, DOCUMENTATION, AUTOMATION



```
$ champagne run --input samplesheet.csv
$ sinclair  run --input samplesheet.csv -params-file ...
```

Consistent usage guidelines and documentation across all pipelines simplifies user training.

GitHub Actions runs CI checks on every change.

Scalable, Reproducible, Sustainable Bioinformatics

NIH Biowulf HPC	GitHub Open Source	Documentation
Secure, Scalable, and high performance computing platform	Collaborate, Contribute, Reuse	Guides, Tutorials, Best Practices

This Poster

Template Repo

Advancing reproducible computational research through high-quality bioinformatics software and data analysis pipelines

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